

10:EM Trauma Airway #1

Section 1: Case Summary

Scenario Title:	10:EM Conference – Trauma Airway #1
Keywords:	Major trauma, Resuscitation, Intubation
Brief Description of Case:	A 53-year old man is brought in after an MVC. He was the unstrained driver and has significant chest trauma from the steering wheel. EMS has attempted needle chest decompression but he remains in severe respiratory distress. The trauma team must assess the patient, identify and treat a tension pneumothorax before safely intubating the patient.

Goals and Objectives	
Educational Goal:	To practice safely securing an airway in a severely traumatized patient
Objectives: (Medical and CRM)	1. Perform a rapid and organized primary survey according to ABCDE (ATLS model) 2. Identify and mitigate sources of shock before performing RSI 3. Vocalize a multi-step RSI plan before proceeding 1. Team leader to give clear, concise directions to team 2. Maintain situational awareness despite competing priorities 3. All team members to use closed loop communication
EPAs Assessed:	N/A

Learners, Setting and Personnel			
Target Learners:	☐ Junior Learners	☒ Senior Learners	☒ Staff
	☒ Physicians	☒ Nurses	☒ RTs
	☐ Other Learners:		
Location:	☒ Sim Lab	☒ In Situ	☐ Other:
Recommended Number of Facilitators:	Instructors: 1		
	Sim Actors: 0-4 (depending on local trauma team structure)		
	Sim Techs: 1		

Scenario Development	
Date of Development:	Sept 2021
Scenario Developer(s):	Dr. Chris Heyd
Affiliations/Institutions(s):	McMaster University
Contact E-mail:	Christopher.heyd@medportal.ca
Last Revision Date:	
Revised By:	
Version Number:	1



10:EM Trauma Airway #1

Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Gerry Gostkowski		Age: 53		Gender: M	Weight: 80 kg
Presenting complaint: Major blunt trauma					
Temp: 36.3	HR: 140	BP: 95/63	RR: 40	O ₂ Sat: 90%	FiO ₂ : NRB
Cap glucose: 10.7			GCS: 13 (E4 V4 M5)		
Triage note: Unrestrained driver in a head-on MVC. Severe respiratory distress. Direct to trauma bay.					
Allergies: Unknown					
Past Medical History: Unknown			Current Medications: Unknown		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, sim actors, SP, etc.)?</i>	
EMS will provide: 53-year old man who was in a head-on MVC. He was unrestrained and was crushed against the steering wheel. Prolonged extrication and found the patient with respiratory distress and left chest injury. We used a needle to decompress his left chest.	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: normal	Neuro: 1 word answers
Resp: tachypneic, reduced air entry L side, no chest rise L side	Head & Neck: C-collar, small forehead lac
Abdo: soft, non-tender	MSK/skin: tender L chest with crepitus
Other: L wrist deformity	

10:EM Trauma Airway #1

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>specify type and whether infant/child/adult</i>)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Intubation supplies Difficult airway cart Chest tube, tray, scalpel, underwater seal
C. Required Medications
RSI medications TXA Blood products
D. Moulage
Blood on forehead, c-collar Decompression needle in L chest
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> Severe respiratory distress with 1 word answers Left chest tenderness and subcutaneous emphysema

10:EM Trauma Airway #1

Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Trauma Team	The team members should be adapted according to the usual trauma response in your emergency department. The team members may be sim participants (in the case of in situ sim or team training) or sim actors (for trauma team leader training).
Bedside RN	Hooks up monitors, places IV lines, draws bloodwork, administers medications, etc.
Charting RN	Completes resuscitation record, assists with blood products, calling x-ray, etc.
Resp Therapist	Prepares intubation equipment, helps with intubation, bagging, ventilation, etc.



10:EM Trauma Airway #1

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: sinus tach HR: 140 BP: 95/63 RR: 40 O ₂ SAT: 90% (NRB) T: 36.3°C GCS: 13 (E4 V4 M5)	<i>Severe respiratory distress. One word answers</i>	<u>Expected Learner Actions</u> ☐ Place on monitors ☐ 2 Large-bore PIVs ☐ Trauma bloodwork ☐ Primary survey (ABCDE) ☐ Vocalize need for intubation ☐ Vocalize need for chest tube ☐ Provide analgesia	<u>Modifiers</u> Needle decompress→ No change <u>Triggers</u> For progression to next state RSI before chest tube or 4 minutes → 2. Worsening Chest tube/thoracostomy → 3. Decompressed	
2. Worsening Rhythm: sinus tach HR: 150 BP: 70/43 RR: 45 O ₂ SAT: 85%	<i>Worsening resp and mental status</i>	<u>Expected Learner Actions</u> ☐ Thoracostomy/chest tube ☐ May give light sedation	<u>Modifiers</u> Needle decompress → No change Paralysis → RR 0 Intubation → No change to Vitals <u>Triggers</u> Chest tube/thoracostomy → 3. Decompressed	
3. Decompressed HR: 120 BP: 105/63 RR: 30 O ₂ SAT: 95%		<u>Expected Learner Actions</u> ☐ E-FAST ☐ May give pRBCs ☐ Vocalize airway plan ☐ Begin RSI	<u>Modifiers</u> <u>Triggers</u> RSI complete → 4. Intubated	Easy intubation with video laryngoscopy
4. Intubated HR: 110 BP: 110/63 RR: 12 (vent) O ₂ SAT: 96%	<i>Paralyzed and ventilated</i>	<u>Expected Learner Actions</u> ☐ Chest/pelvis XR ☐ Arrange CT Scan ☐ Chest tube to underwater seal ☐ Log roll ☐ Call ICU/Surgery	<u>Modifiers</u> <u>Triggers</u> All action complete→ End Case	



10:EM Trauma Airway #1

Appendix A: Laboratory Results

No lab results given in this case	
-----------------------------------	--

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!

No X-rays given

E-FAST

R Lung – Normal

L Lung – No lung sliding

Epigastrium – No pericardial effusion

Abdomen – No free fluid

10:EM Trauma Airway #1

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

1. Use A-B-C-D-E to quickly assess the traumatized patient

- Identify the need for intubation and anatomic and physiologic challenges
- (Logroll can be deferred until intubation is complete)

2. Resuscitate before you intubation. Identify causes of shock that will cause decompensation during rapid-sequence intubation. Correct these in physiologic order (not ABCDE order).

Causes of shock in traumatized patients

- Obstructive: pericardial tamponade, tension pneumothorax
- Hypovolemic: internal or external hemorrhage
- Distributive: Neurogenic shock (diagnosis of exclusion)
- Medical: Intoxication/withdrawal, medication-related, underlying medical issue (based on history)

3. Develop and vocalize a multi-step airway plan.

- Expect an anatomically difficult airway
- Prepare adjuncts and back-up plans
- Prepare for surgical airway (at minimum, mental rehearsal)

References

1. Petrosoniak A & Hicks C. (2018) Resuscitation Resequenced: A Rational Approach to Patients with Trauma in Shock. *Emerg Med Clin N Am* 36:41–60. <http://dx.doi.org/10.1016/j.emc.2017.08.005>
2. American College of Surgeons. (2018). *Advanced Trauma Life Support: Student Course Manual*, 10th Ed. American College of Surgeons, Chicago, IL.

